

HDOC Day-24

C Programming Assignment — Decreasing Patterns

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Program	B.Tech CSE

Objective

Write a menu-driven C program using a switch statement to print the selected decreasing pattern. Use nested loops without arrays. Each pattern is implemented as a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display the menu containing the four decreasing-pattern choices.
3. Read the user's choice and the required number of rows.
4. Use a switch statement to call the selected pattern function.
5. Use nested loops to print the required decreasing pattern.
6. Display the output and stop.

C Program

The following program provides all four activities through a switch-based menu.

```
#include <stdio.h>

int n;

void decreasingRightTriangle()
{
    for (int i = n; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

void decreasingNaturalNumber()
{
    for (int i = n; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingRepeatedNumber()
{
    for (int i = n; i >= 1; i--)
    {
        for (int j = 1; j <= i; j++)
            printf("%d ", i);
        printf("\n");
    }
}

void reverseNumberTriangle()
{
    for (int i = n; i >= 1; i--)
    {
        for (int j = i; j >= 1; j--)
            printf("%d ", j);
        printf("\n");
    }
}

int main()
{
    int choice;

    printf("HDOC Day-24\n");
    printf("1. Decreasing right-angled triangle\n");
    printf("2. Decreasing natural-number pattern\n");
    printf("3. Decreasing repeated-number pattern\n");
    printf("4. Reverse-number triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("Enter number of rows: ");
            scanf("%d", &n);
            decreasingRightTriangle();
            break;

        case 2:
            printf("Enter number of rows: ");
            scanf("%d", &n);
            decreasingNaturalNumber();
            break;
```

```
    case 3:
        printf("Enter number of rows: ");
        scanf("%d", &n);
        decreasingRepeatedNumber();
        break;

    case 4:
        printf("Enter number of rows: ");
        scanf("%d", &n);
        reverseNumberTriangle();
        break;

    default:
        printf("Invalid choice.\n");
}

return 0;
}
```

1. Decreasing right-angled triangle

Sample input:

Enter number of rows: 4

Output:

```
* * * *
* * *
* *
*
```

2. Decreasing natural-number pattern

Sample input:

Enter number of rows: 4

Output:

```
1 2 3 4
1 2 3
1 2
1
```

3. Decreasing repeated-number pattern

Sample input:

Enter number of rows: 4

Output:

```
4 4 4 4
3 3 3
2 2
1
```

4. Reverse-number triangle

Sample input:

Enter number of rows: 4

Output:

```
4 3 2 1
3 2 1
2 1
1
```

Test-Case Table

TC	Choice	Input	Expected output / result
1	1	n = 4	4, 3, 2 and 1 stars in successive rows
2	2	n = 4	1-4, 1-3, 1-2 and 1 in successive rows
3	3	n = 4	4 repeated 4 times, then 3, 2 and 1
4	4	n = 4	4 3 2 1, then 3 2 1, then 2 1, then 1
5	9	—	Invalid choice message

Conclusion

The program successfully uses a switch statement and nested loops to generate the four required decreasing patterns without using arrays.

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